MCESAN: A Modality Complementary Enhancement and Sparse Attention Network for Joint Hyperspectral and LiDAR Image Classification

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***Abstract*—Hyperspectral images (HSIs), with their hundreds of contiguous narrow bands, capture detailed physicochemical properties of land covers; LiDAR provides high-precision 3-D spatial and geometric structure, and the two modalities are highly complementary. However, differences in imaging mechanisms lead to insufficient cross-modal feature consistency, and conventional self-attention mechanisms tend to introduce numerous redundant dependencies in long-range modeling, impairing fusion accuracy. To address this, we propose a modality complementary enhancement and sparse attention network (MCESAN) for joint classification. First, we construct a modality complementary enhancement and correction module, composed of an adaptive discrepancy perception unit and a global consistency correction unit, to achieve precise feature alignment and semantic calibration. Then, we design a selective key sparse attention (SKSA) mechanism that uses LiDAR spatial features as queries and HSI spectral features as keys and values to establish a directional interaction framework, suppressing redundant associations and focusing on critical cross-modal dependencies. Experiments on the Houston2013, Trento, and MUUFL datasets demonstrate overall accuracy (OA) superior to mainstream methods, validating its effectiveness and robustness in complex scenes.**

***Index Terms*—Attention mechanism, cross-modal fusion, hyperspectral and LiDAR, long-range sparse modeling, modality complementary enhancement.**

I. Introduction

HYPERSPECTRAL image (HSI) and LiDAR data are two highly complementary data sources in remote sensing land-cover classification [1]. HSI has hundreds of continuous narrow bands. It can finely reflect the physical and chemical properties of ground objects [2], [3], [4]. It plays a key role in mineral exploration [5], agricultural monitoring [6], and environmental assessment [7]. LiDAR provides high-precision 3-D spatial information [8], [9], [10]. It can compensate for the lack of elevation representation in HSI [11]. However, relying solely on HSI spectral information makes it difficult to distinguish land covers with similar spectra but different spatial structures, while relying solely on LiDAR spatial information fails to identify categories with significant spectral differences but similar elevations. Both modalities have inherent limitations when used individually.

The spectral discrimination advantage of HSI and the spatial structure advantage of LiDAR complement each other. Their complementarity provides core data support for improving the accuracy and reliability of land-cover classification in complex scenes [12], [13], [14]. Therefore, how to effectively fuse heterogeneous features has become an important research direction in intelligent remote sensing interpretation [15].

In recent years, deep learning has become the mainstream technical route for joint HSI and LiDAR classification. This is due to its powerful feature learning capability. Early dual-branch CNN architectures, such as Coupled CNNs [11], IP-CNN [16], and Two-Stream CNN [8], initially achieved collaborative modeling of spectral and elevation features. Subsequently, Joint CNNs [17] introduced 3-D CNN and morphological modules. M2FNet [18] proposed a multiscale 3-D–2-D hybrid CNN. AMSSE-Net [19] and DSHFNet [20] improved representation capability through adaptive fusion and dynamic hierarchical interaction, respectively. MCFNet [21] and MADNet [22] strengthened collaborative representation through multiscale cross-domain fusion and dynamic scale interaction. However, the inherent local receptive field of CNNs limits their effective capture of long-range dependencies [23].

To compensate for these shortcomings, Transformer-based methods have gradually become a research hotspot. They leverage the advantage of self-attention mechanisms in global modeling. MTNet [24] mines long-range dependencies through spectral and spatial Transformers. Cross-HL [25] constructs asymmetric interaction with LiDAR as query and HSI as key-value. FrIT [26], LIIT [27], and S2EFT [28] improve the architecture from the perspectives of fractional Fourier transform, convolution-embedded attention, and spatial–spectral–elevation joint modeling, respectively. CAB-HL [29] explores cross-attention bimodal collaboration. MCFT [30] proposes a multimodal contrastive fusion Transformer. CEAT [31] proposes a cross-modal enhanced interactive Transformer. FETNet [32] introduces fuzzy logic into the self-attention mechanism to model class uncertainty and fuzzy boundaries. However, pure Transformer architectures are

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relatively insufficient in mining local detailed features. They also suffer from training instability [33].

To balance local details and global associations, CNN-Transformer hybrid architectures have been further proposed. GLT-Net [34] first uses CNN to extract local features and then uses Transformer to model spectra. HCTNet [35] designs a hierarchical dual-branch with cross-token attention. DF2NCECT [36] and MS2CANet [37] strengthen hybrid architectures through semantic bias mitigation and multiscale attention, respectively. CITNet [38] improves fineness through convolution–self-attention coupling. DCTN [39] explores a dual-branch convolutional Transformer with efficient interactive self-attention for spectral-spatial feature modeling. PGPL [40] introduces a few-shot prototype-guided progressive learning method that achieves effective classification with limited labeled samples. IACF-Net [41] and CPFENet [42] propose interactive adaptive cross-fusion and central pixel feature enhancement, respectively. DFNet [43], MSAFNet [44], and AF2DN [45] explore deep fuzzy fusion, multiscale adaptive fusion, and attention-guided frequency decomposition, respectively. MFNet [46], HCAFNet [47], and S2GCCFN [48] design hybrid attention multi-level fusion, hierarchical cross-modal attention, and shared-specific graph-guided fusion, respectively. HCGFN [49], CGMFN [50], and GASSF-Net [51] propose heterogeneous contrastive graph fusion, CNN-GCN collaborative multimodal frequency network, and geometric algebra-driven fusion, respectively. MCAFNet [52], SA-FCM [53], GFSFN [54], and MDAGNet [55] construct multiscale cross-attention, subpixel attention with frequency domain cross-modulation, multilevel feature gating fusion, and multilevel difference aggregation gating discriminative network, respectively.

Although the above hybrid architectures have made significant progress, they still generally have limitations in three aspects. First, HSI and LiDAR suffer from modality misalignment due to different imaging mechanisms, yet most existing methods directly concatenate or simply fuse features without effective per-pixel difference perception and local spatial correction before fusion [56], [57], [58]. Second, existing methods lack global semantic calibration from the channel dimension, leaving noisy channels unsuppressed and key semantic channels unenhanced, which degrades subsequent cross-modal modeling [59], [60]. Third, traditional self-attention indiscriminately computes all associations, introducing numerous low-contribution redundant associations that dilute key cross-modal discriminative information and limit fine-grained classification accuracy [61], [62].

Specifically, among the representative cross-modal attention methods, Cross-HL [25] constructs asymmetric interaction with LiDAR as query and HSI as key/value but directly applies attention to raw features without explicit modality discrepancy modeling. CAB-HL [29] employs bidirectional cross-attention for bimodal collaboration yet relies on simple feature concatenation without spatial misalignment correction. CEAT [31] introduces cross-modal enhanced interactive attention but lacks dedicated modules for per-pixel discrepancy perception or global semantic calibration.

To address these three challenges, this article proposes a joint classification method for HSI and LiDAR images based on modality complementary enhancement and sparse long-range modeling. The solutions are as follows.

1) To address the insufficient precise alignment of bimodal features at the local spatial level, an adaptive difference perception module within the modality complementary enhancement correction (MCEC) module is proposed. It quantifies per-pixel differences and generates cross-modal spatial weights, achieving differentiated local feature correction. Unlike geometric registration methods that estimate explicit spatial offsets, adaptive discrepancy perception module (ADPM) performs feature-level response correction by modulating each modality's representation based on local discrepancy, without altering the spatial coordinates or requiring image resampling.
2) To address the missing distribution calibration of bimodal features at the global semantic level, a global consistency correction module within the MCEC module is proposed. It uses a multipooling combination strategy of global average pooling, max pooling, and standard deviation pooling to comprehensively characterize the channel semantic distribution. Guided by cross-modal channel weights, it achieves global semantic calibration. It suppresses noisy channels and strengthens effective semantic channels. This improves the consistency of bimodal features in global semantic expression.
3) To address the redundant dependency problem of traditional self-attention in long-range modeling, a selective key sparse attention (SKSA) mechanism is proposed. A cross-modal directed interaction framework is constructed with LiDAR spatial features as queries and HSI spectral features as keys. This allows spatial structure to explicitly guide the modeling of spectral long-range dependencies. Meanwhile, a sparse selection operator is introduced to adaptively refine the attention similarity matrix. It removes low-contribution redundant associations and precisely focuses on key cross-modal dependency relationships.

II. Proposed Method

The proposed modality complementary enhancement and sparse attention network (MCESAN), a joint HSI–LiDAR classification network based on modality complementary enhancement and long-range sparse modeling, is designed to achieve precise cross-modal feature fusion and efficient global modeling in complex land-cover scenes. Its overall architecture is shown in Fig. 1 The network primarily consists of specialized feature extraction branches, an MCEC module, and an SKSA mechanism. MCEC further comprises an ADPM and a global consistency calibration module (GCCM), which sequentially perform collaborative correction and enhancement on the dual-modal features at the local spatial level and the global channel level, respectively.

A. Specialized Feature Extraction Branches

To accommodate the intrinsic data characteristic differences between HSI and LiDAR, the network employs dedicated feature extraction branches at the input. Let the original HSI data be a 3-D data cube encompassing spatial and spectral dimensions; the LiDAR data mainly provide elevation and spatial structure information, with spatial dimensions matching those of the HSI.

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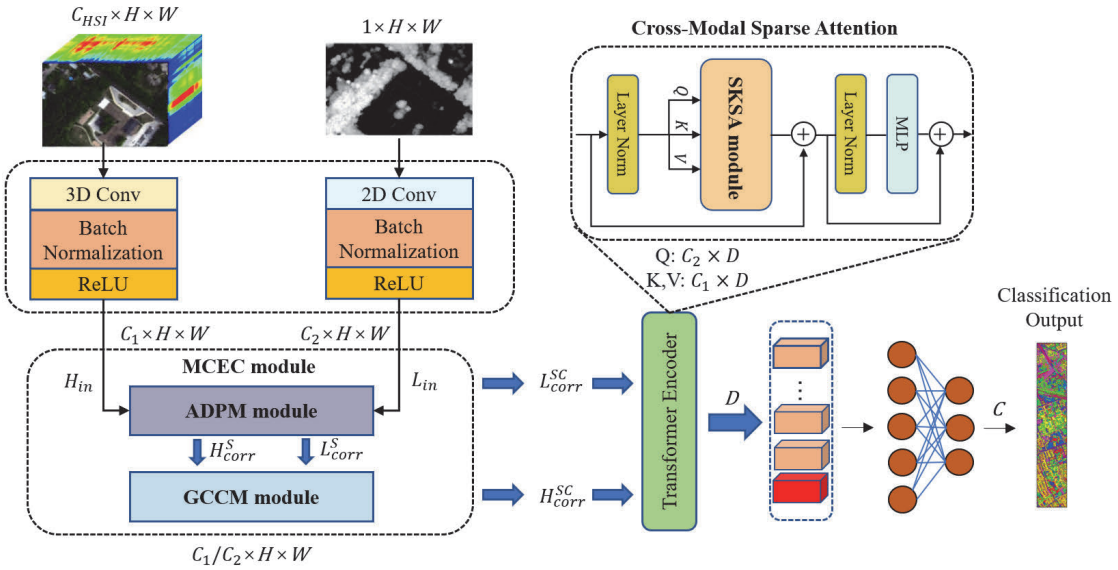


Fig. 1. Architecture of MCESAN.

For HSI data, 2-D convolution would sever the inherent correlation between the spectral and spatial dimensions, leading to a loss of spectral discriminative information. Hence, 3-D convolution is adopted for joint spectral–spatial feature extraction, preserving both spectral response characteristics and spatial context within a single convolution operation. Given an input local neighborhood patch H_{patch}, it passes through a feature extraction layer consisting of 3-D convolution, batch normalization, and ReLU activation, yielding the initial HSI feature H_{in}

$$H_{\text{in}} = \sigma\left(\text{BN}\left(\text{Conv3D}\left(H_{\text{patch}}\right)\right)\right) \tag{1}$$

where $\sigma(\cdot)$ denotes the ReLU activation function and BN$(\cdot)$ denotes batch normalization.

For LiDAR data, which primarily carries elevation values and spatial structure with relatively low feature dimensionality, a lightweight 2-D convolution module is employed for spatial feature enhancement. Similarly, a local neighborhood patch L_{patch} serves as input, and after processing by 2-D convolution, batch normalization, and ReLU activation, the initial LiDAR feature L_{in} is obtained

$$L_{\text{in}} = \sigma\left(\text{BN}\left(\text{Conv2D}\left(L_{\text{patch}}\right)\right)\right). \tag{2}$$

After extraction by these specialized branches, the initial HSI feature $H_{\text{in}} \in \mathbb{R}^{W\times H\times C_1}$ and LiDAR feature $L_{\text{in}} \in \mathbb{R}^{W\times H\times C_2}$ share matching spatial dimensions, providing a unified input format for the subsequent modality complementary enhancement and correction layer.

B. MCEC Module

Owing to marked differences in imaging mechanisms, HSI and LiDAR features suffer from response shifts and misalignments at the local spatial level and from uneven channel responses and noise interference at the global semantic level. Most existing methods directly concatenate heterogeneous features or apply superficial attention fusion, lacking explicit modeling and correction of intermodal discrepancies, and thus fail to achieve precise feature-level alignment. To address this, we design an MCEC module, which consists of an ADPM and a GCCM connected in series. ADPM performs discrepancy perception and pixel-wise correction at the local spatial level, while GCCM constrains and calibrates the semantic distribution at the global channel level. Working in synergy, they effectively mitigate the adverse effects of modality gaps and noise on subsequent fusion modeling while preserving the discriminative information of each modality.

1) Adaptive Discrepancy Perception Module (ADPM): The core idea of ADPM is to quantify the discrepancy between HSI and LiDAR features at each spatial position and generate adaptive correction weights based on the discrepancy magnitude, thereby achieving fine-grained feature correction in local regions. ADPM performs feature-level response correction, rather than geometric registration. It does not estimate explicit spatial offsets or resample feature maps. Instead, it computes per-pixel discrepancy as an indicator of local response mismatch and uses cross-modal compensation to adaptively adjust feature representations. Its structure is shown in Fig. 2.

HSI features capture spectral reflectance characteristics, whereas LiDAR features represent elevation distributions; their orders of magnitude differ considerably. Directly computing differences would allow numerical ranges to dominate the result, failing to reflect genuine information matching. Hence, we first apply channel-wise normalization to the input features H_{in} and L_{in} to eliminate magnitude discrepancies. Let c denote the channel index; the normalization for the cth channel is given by

$$\hat{H}^{(c)}_{in}(x,y) = \frac{H^{(c)}_{in}(x,y) - \min\limits_{x,y} H^{(c)}_{in}(x,y)}{\max\limits_{x,y} H^{(c)}_{in}(x,y) - \min\limits_{x,y} H^{(c)}_{in}(x,y) + \epsilon}$$
$$\hat{L}^{(c)}_{in}(x,y) = \frac{L^{(c)}_{in}(x,y) - \min\limits_{x,y} L^{(c)}_{in}(x,y)}{\max\limits_{x,y} L^{(c)}_{in}(x,y) - \min\limits_{x,y} L^{(c)}_{in}(x,y) + \epsilon} \tag{3}$$

where ϵ is a small constant to prevent division by zero. After normalization, both features lie in a common numerical range, ensuring that subsequent discrepancy computation is not biased by the original magnitude scales.

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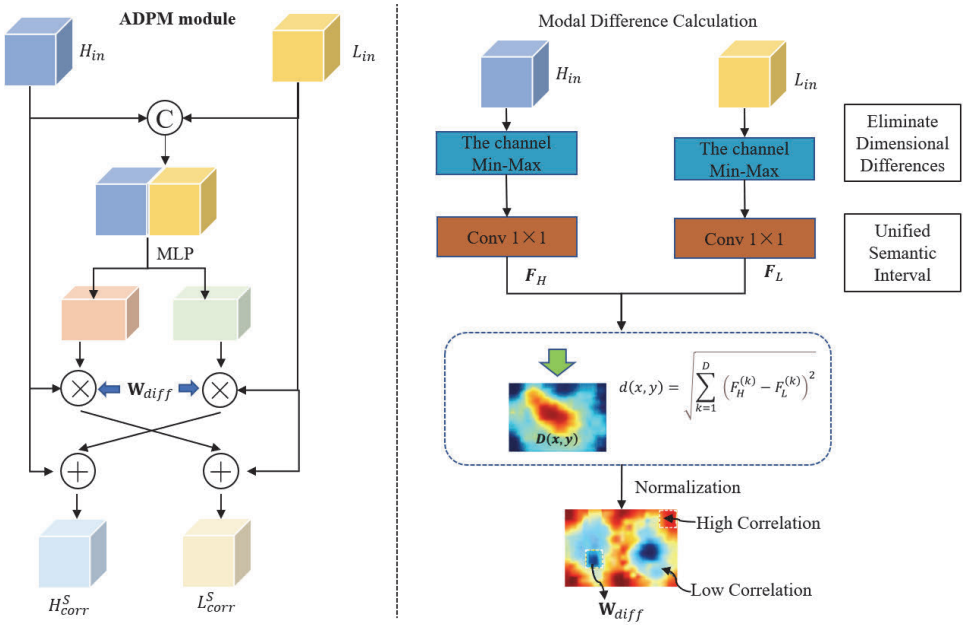


Fig. 2. Structure of the adaptive discrepancy perception module.

Although normalization removes magnitude differences, HSI and LiDAR features still reside in distinct physical semantic spaces. To measure discrepancy on a unified semantic scale, we further project the normalized features into a shared semantic embedding space via linear mappings

$$F_H = \phi\left(\hat{H}_{in}\right) \in \mathbb{R}^{W\times H\times D}$$
$$F_L = \psi\left(\hat{L}_{in}\right) \in \mathbb{R}^{W\times H\times D} \tag{4}$$

where $\phi(\cdot)$ and $\psi(\cdot)$ are implemented by 1×1 convolutions and D is the embedding dimension.

The channel-wise normalization in (3) addresses the numerical range discrepancy between the two modalities, ensuring that subsequent computations are not dominated by magnitude differences. The linear projection in (4), in contrast, addresses a different issue—HSI spectral semantics and LiDAR geometric semantics reside in inherently distinct physical spaces, and projecting them into a shared embedding space establishes a semantic correspondence in a higher-dimensional abstract space, enabling meaningful cross-modal discrepancy measurement. After this step, the cross-modal features reside in the same metric space, so the computed discrepancy carries clear semantic meaning.

We then compute the pixel-wise Euclidean distance between the two projected features to obtain a modality discrepancy map $D \in \mathbb{R}^{W\times H}$, quantifying the local mismatch. Larger discrepancy values indicate lower feature correspondence at that location, calling for stronger cross-modal compensation. The discrepancy map is defined as

$$\mathbf{D}(x,y) = \|F_H(x,y) - F_L(x,y)\|_2 = \sqrt{\sum_{k=1}^{D}\left(F_H^{(k)}(x,y) - F_L^{(k)}(x,y)\right)^2}. \tag{5}$$

The discrepancy map is normalized to obtain modality discrepancy weights $W_{\text{diff}} \in \mathbb{R}^{W\times H}$

$$\mathbf{W}_{diff}(x,y) = \frac{\mathbf{D}(x,y) - min(\mathbf{D})}{max(\mathbf{D}) - min(\mathbf{D}) + \epsilon}. \tag{6}$$

To establish spatial complementary relationships between modalities, we concatenate the input features H_{in} and L_{in} along the channel dimension, yielding a fused feature F_C that allows the subsequent weight generation network to perceive local information from both modalities

$$F_C = \text{Concat}(H_{in}, L_{in}). \tag{7}$$

A multilayer perceptron (MLP) then nonlinearly maps the fused feature to capture more complex spatial complementary relations between the two modalities. The output is passed through a sigmoid activation to confine values to [0, 1] and then split into two spatial weight maps, W_H^S and W_L^S, corresponding to the HSI and LiDAR branches, respectively,

$$F_{MLP} = \text{MLP}(F_C) \tag{8}$$
$$W_H^S, W_L^S = \text{Split}\left(\text{Sigmoid}\left(F_{MLP}\right)\right). \tag{9}$$

With the discrepancy weight and spatial weights, we establish a bidirectional residual local correction mechanism. The correction of HSI features is guided by the spatial weight generated from the LiDAR branch, and vice versa, enabling the two modalities to reference and complement each other within local regions. The correction increments are formulated as

$$\Delta H = W_L^S \cdot \mathbf{W}_{diff} \cdot L_{in}$$
$$\Delta L = W_H^S \cdot \mathbf{W}_{\text{diff}} \cdot H_{\text{in}}. \tag{10}$$

ADPM uses the discrepancy map as a relative indicator rather than assuming all large discrepancies correspond to misalignment. The discrepancy map is computed in a shared embedding space, suppressing magnitude-level differences and highlighting semantic mismatches. The compensation is applied via residual connections and controlled by spatial weights, preventing over-compensation. Meanwhile, GCCM suppresses channel noise, and SKSA focuses on discriminative features through sparse attention. Therefore, ADPM

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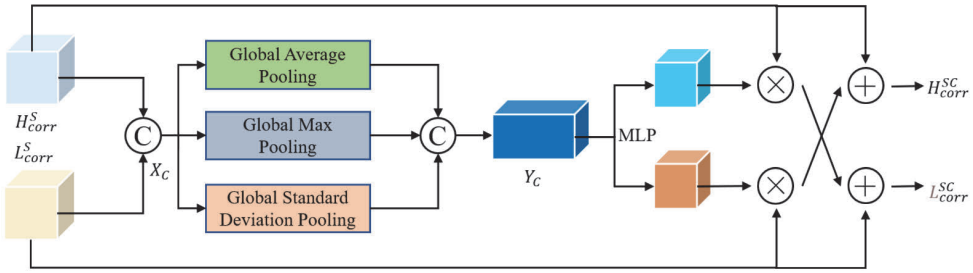


Fig. 3. Structure of the global consistency calibration module.

provides an adaptive guidance signal while the overall network preserves robustness against noise and modality-specific information.

Finally, the correction increments are added to the original features in a residual manner, yielding the spatially corrected features H^S_{corr} and L^S_{corr}

$$H^S_{\text{corr}} = H_{\text{in}} + \Delta H$$
$$L^S_{\text{corr}} = L_{in} + \Delta L. \tag{11}$$

The residual formulation, instead of direct replacement, preserves the discriminative information in the original features while using cross-modal compensation to correct locally misaligned regions, striking a balance between alignment accuracy and information fidelity.

2) Global Consistency Calibration Module (GCCM): After ADPM, local representation deviations are largely mitigated, but from a global semantic perspective, issues such as uneven channel response strengths, inconsistent semantic distributions, and noisy channel interference may still remain.

Therefore, a GCCM is introduced on top of the local correction to further calibrate the semantic distributions at the global channel level. Its structure is shown in Fig. 3.

First, the spatially corrected features H^S_{corr} and L^S_{corr} from ADPM are concatenated along the channel dimension to form a unified representation X_C, allowing subsequent global pooling to simultaneously capture channel-wise semantic information from both modalities

$$X_C = \text{Concat}\left(H^S_{\text{cort}}, L^S_{\text{corr}}\right). \tag{12}$$

To fully characterize the global semantic distribution, we compress X_C using a combined strategy of global average pooling, global max pooling, and global standard deviation pooling. Global average pooling extracts the mean response across the spatial extent, $\text{Avg}(X_C)$, reflecting the overall activation level of each channel; global max pooling retains the most salient responses, $\text{Max}(X_C)$, capturing the peak discriminative activations; and global standard deviation pooling reflects the spatial variability and dispersion of channel responses, $\text{Std}(X_C)$. These three statistics—representing average magnitude, peak saliency, and spatial dispersion—collectively describe the channel-wise feature distribution, providing more comprehensive information for subsequent channel weight learning. The three pooling results are concatenated along the channel dimension to form a global channel descriptor Y_C

$$Y_C = \text{Concat}\left(\text{Avg}\left(X_C\right), \text{Max}\left(X_C\right), \text{Std}\left(X_C\right)\right). \tag{13}$$

The descriptor Y_C is fed into an MLP for nonlinear mapping, learning the importance of each channel in the global semantic representation. The output is squashed to [0, 1] by a sigmoid function and split into channel weight vectors W^C_H and W^C_L for the HSI and LiDAR branches, respectively,

$$Y_{MLP} = \text{MLP}\left(Y_C\right) \tag{14}$$
$$W^C_H, W^C_L = \text{Split}\left(\text{Sigmoid}\left(Y_{MLP}\right)\right). \tag{15}$$

The channel weights for the HSI and LiDAR branches are derived from the pooled statistics of both modalities, enabling cross-guidance because each modality receives correction weights estimated from the joint distribution rather than from its own statistics alone.

Finally, a cross-modal channel compensation mechanism is applied to achieve global semantic calibration. The HSI feature is corrected under the guidance of the LiDAR-generated channel weights, and symmetrically, the LiDAR feature is corrected by the HSI-generated channel weights. This cross-guidance further enhances the consistency of dual-modal global semantic representations

$$H^{SC}_{corr} = H^S_{corr} + W^C_L \cdot L^S_{corr}$$
$$L^{SC}_{\text{corr}} = L^S_{\text{corr}} + W^C_H \cdot H^S_{\text{corr}}. \tag{16}$$

After GCCM processing, the dual-modal features not only preserve the structural complementarity achieved in ADPM but also attain more coordinated channel representations in the overall semantic distribution. The serial collaboration of ADPM and GCCM mitigates modality discrepancies and noise from both local spatial and global channel perspectives, providing a high-quality, highly consistent multimodal feature foundation for subsequent cross-modal long-range feature modeling.

C. SKSA Mechanism

After processing by the MCEC module, the complementarity between HSI spectral features and LiDAR spatial–geometric features is significantly strengthened, while redundant information and intermodal noise are preliminarily suppressed. On this basis, it is necessary to further model long-range dependencies between cross-modal features.

However, conventional Transformer self-attention has two limitations in dual-modal scenarios: first, multimodal interaction adopts a symmetric homogeneous pattern, failing to fully exploit the inherent differences between HSI spectral discrimination and LiDAR spatial constraints; second, indiscriminate dense computation over all feature relations tends to introduce numerous low-contribution redundant associations and invalid mismatches, diluting attention weights on critical cross-modal dependencies.

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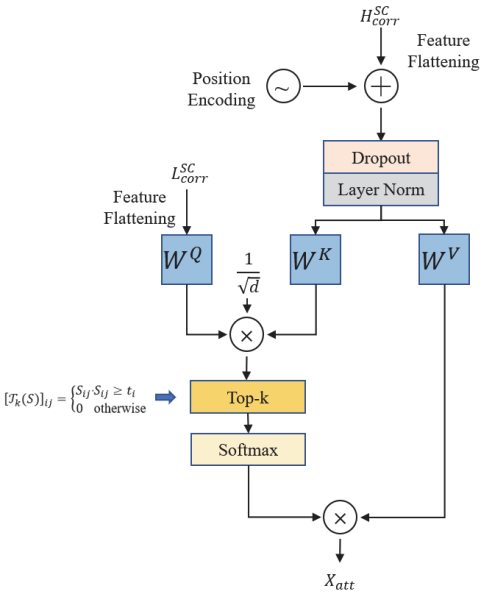


Fig. 4. Structure of the SKSA mechanism.

To overcome these limitations, this section proposes an SKSA mechanism to replace the conventional self-attention in Transformers. Its structure is illustrated in Fig. 4.

SKSA constructs an asymmetric cross-modal directional interaction framework tailored to the information characteristics of HSI and LiDAR. LiDAR data provide geometric information such as elevation, morphology, and spatial distribution, making them suitable for a spatial guidance role in cross-modal association; HSI data contain rich, contiguous spectral information, ideal for spectral feature matching and class discrimination. Consequently, the corrected LiDAR features form the query vectors Q, while the corrected HSI features form the key vectors K and value vectors V

$$\begin{aligned} Q &= W^Q L_{\text{corr}}^{SC} \\ K &= W^K H_{\text{corr}}^{SC} \\ V &= W^V H_{\text{corr}}^{SC} \end{aligned} \tag{17}$$

where W^Q, W^K, and W^V are linear projection matrices for query, key, and value. This asymmetric construction enables LiDAR spatial features to act as the querying guide in cross-modal interaction, while HSI spectral features serve as the queried carrier for spectral matching and feature aggregation.

After obtaining Q and K, a cross-modal attention similarity matrix S is computed to quantify the interaction strength between modalities

$$S = \frac{Q \cdot K^{\mathrm{T}}}{\lambda} \tag{18}$$

where $\lambda = \sqrt{d}$ is a scaling factor and d is the feature dimension of the query and key vectors. The scaling factor regulates the numerical range, preventing gradient vanishing in the subsequent softmax. Each element S_{ij} represents the association strength between the ith LiDAR spatial position and the jth HSI spectral dimension.

To prevent the dilution of attention by low-contribution redundant associations within the dense similarity matrix, a sparse selection operator $T_k(\cdot)$ is introduced to adaptively refine S

$$[\mathrm{T}_k(S)]_{ij} = \begin{cases} S_{ij}, & S_{ij} \geq t_i \\ 0, & otherwise \end{cases} \tag{19}$$

where t_i is the $\lceil k \cdot N \rceil$th largest value in the ith row of S, serving as the row-wise threshold, with N being the total number of elements in that row and $k \in (0, 1]$ representing the proportion of retained elements. The sparsity parameter k is adjustable and can be tuned according to the land-cover complexity of the dataset.

After sparse selection, softmax normalization is applied to obtain the sparse attention weight matrix $\hat{S}$

$$\hat{S} = \mathrm{softmax}\left(\mathrm{T}_k(S)\right). \tag{20}$$

Multiplying $\hat{S}$ with the value vectors V yields the aggregated cross-modal feature X_{att}

$$X_{att} = \hat{S} \cdot V. \tag{21}$$

This fused feature preserves the core discriminative spectral information of HSI while incorporating LiDAR spatial constraints; simultaneously, redundant associations and noise are eliminated through sparse screening.

SKSA is embedded into the Transformer encoder in place of the original self-attention, constructing an improved Transformer for joint HSI–LiDAR classification. Taking the corrected features H_{corr}^{SC} and L_{corr}^{SC} as input, the core encoder operations are

$$x_l' = H_{\text{corr}}^{SC} + \mathrm{SKSA}\left(H_{\text{corr}}^{SC}, L_{\text{corr}}^{SC}\right) \tag{22}$$

$$x_l = x_l' + \mathrm{MLP}\left(\mathrm{LN}\left(x_l'\right)\right) \tag{23}$$

where $x_l = [x_{\text{cls}}, x_1, \ldots, x_d]$ is the full output sequence, and x_{cls} is a learnable class token that aggregates high-level semantic information and cross-modal complementary discriminative features from the fused representation. Finally, x_{cls} is passed through a linear transformation and a softmax function to produce the classification probabilities

$$F = \mathrm{Softmax}\left(\mathrm{Linear}\left(x_{cls}\right)\right). \tag{24}$$

Through the above design, SKSA enhances the quality of cross-modal long-range dependency modeling in two aspects: interaction structure and attention screening. At the interaction structure level, the asymmetric formulation with LiDAR as query and HSI as key/value allows spatial structure information to exert a guiding role while spectral features carry out discriminative representation. At the attention screening level, the sparse selection operator refines critical cross-modal associations via adaptive thresholding, effectively suppressing the dilution of attention weights by redundant information, thereby improving the precision of feature aggregation while reducing computational cost.

D. Loss Function

In the joint HSI–LiDAR classification task, the training objective is to minimize the discrepancy between predicted class distributions and the true class labels. Let the training set consist of N samples and the total number of land-cover classes be C. For the ith sample, the ground-truth label is $y_i \in \{1, 2, \ldots, C\}$, and the predicted class probability vector is

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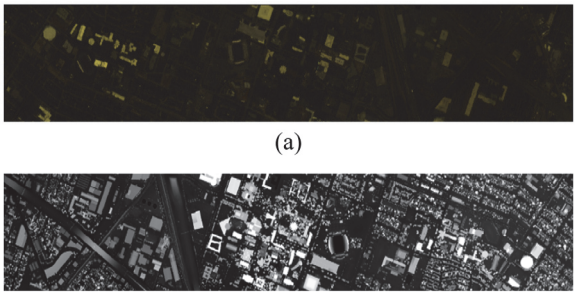


Fig. 5. Houston2013 dataset. (a) False-color image for HS. (b) Grayscale image for LiDAR.

$\hat{y}_i = [\hat{y}_i^1, \hat{y}_i^2, \ldots, \hat{y}_i^C]^\top$. We adopt the cross-entropy loss, which is widely used in remote sensing image classification

$$L_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_i^c \log\left(\hat{y}_i^c\right) \tag{25}$$

where y_i^c is the ground-truth label of the ith sample for class c (in one-hot encoding), and $\hat{y}_i^c$ is obtained from the softmax output as in (24). During training, the Adam optimizer is employed for end-to-end joint optimization of all network parameters, including the specialized feature extraction branches, the MCEC module, and the SKSA mechanism. Minimizing the cross-entropy loss drives the model to progressively improve classification accuracy.

III. Experimental Results

To comprehensively evaluate the effectiveness of the proposed method, experiments are conducted on three internationally recognized HSI–LiDAR dual-source benchmark datasets: Houston2013, Trento, and MUUFL. The three datasets differ in scene complexity, number of land-cover classes, and sample distribution, thus allowing a thorough examination of the model's classification performance and generalization ability under varied conditions.

A. Datasets

1) *Houston2013 Dataset:* This dataset was acquired over the University of Houston and its surrounding urban area. The image is affected by cloud cover and has a complex scene. It contains 15 land-cover classes. The HSI was acquired by the ITRES CASI-1500 sensor. It has 144 bands with a wavelength range of 364–1046 nm and a spatial resolution of 1 m. The LiDAR provides elevation information for the corresponding area. The training and test sample splits are shown in Table I. Fig. 5 presents the visualization of this dataset, including a false-color image of the HSI and a grayscale image of the LiDAR.
2) *Trento Dataset:* This dataset was acquired over the suburban area of Trento, Italy. It exhibits a clear urban–rural transitional character and contains six land-cover classes. The HSI was acquired by the AISA Eagle sensor. It has 63 bands with a wavelength range of 0.42–0.99 μm and a spatial resolution of 1 m. The LiDAR was acquired by the Optech ALTM3100EA sensor. The sample splits are shown in Table II. Fig. 6 presents the visualization of this dataset, including a false-color image of the HSI and a grayscale image of the LiDAR.
3) *MUUFL Dataset:* This dataset was acquired over Starkville, Mississippi, USA. It features dense urban man-made objects and contains 11 land-cover classes. The HSI has 64 bands with a wavelength range of 375–1050 nm and a spatial resolution of 0.54 m. The LiDAR provides high-resolution elevation information. The sample splits are shown in Table III. Fig. 7 presents

TABLE I

Number of Training and Test Samples for Each Class in the Houston2013 Dataset

Class No.	Class Name	Training Samples	Test Samples
C1	Healthy Grass	198	1053
C2	Stressed Grass	190	1064
C3	Synthetic Grass	192	505
C4	Trees	188	1056
C5	Soil	186	1056
C6	Water	182	143
C7	Residential	196	1072
C8	Commercial	191	1053
C9	Road	193	1059
C10	Highway	191	1036
C11	Railway	181	1054
C12	Parking Lot 1	192	1041
C13	Parking Lot 2	184	285
C14	Tennis Court	181	247
C15	Running Track	187	473
Total		2832	12197

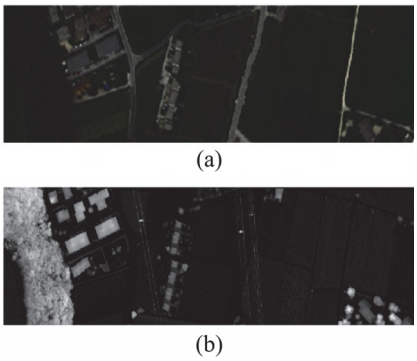


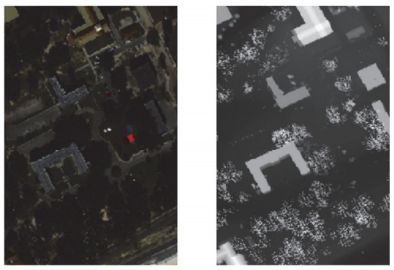
Fig. 6. Trento dataset. (a) False-color image for HS. (b) Grayscale image for LiDAR.

TABLE II

Number of Training and Test Samples for Each Class in the Trento Dataset

Class No.	Class Name	Training Samples	Test Samples
C1	Apple Trees	129	3905
C2	Buildings	125	2778
C3	Ground	105	374
C4	Woods	154	8969
C5	Vineyard	184	10317
C6	Roads	122	3052
Total		819	29395

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(a) (b)

Fig. 7. MUUFL dataset. (a) False-color image for HS. (b) Grayscale image for LiDAR.

TABLE III
NUMBER OF TRAINING AND TEST SAMPLES FOR EACH CLASS IN THE MUUFL DATASET

Class No.	Class Name	Training Samples	Test Samples
C1	Trees	150	23096
C2	Mostly Grass	150	4120
C3	Mixed Ground Surface	150	6732
C4	Dirt and Sand	150	1676
C5	Road	150	6537
C6	Water	150	316
C7	Building Shadow	150	2083
C8	Building	150	6090
C9	Sidewalk	150	1235
C10	Yellow Curb	150	33
C11	Cloth Panels	150	119
Total		1650	52037

the visualization of this dataset, including a false-color image of the HSI and a grayscale image of the LiDAR.

B. Experimental Setup

Three core metrics commonly used in the field of remote sensing classification are adopted to evaluate classification performance: overall accuracy (OA), average accuracy (AA), and the Kappa coefficient.

1) *Evaluation Metrics:* Three core metrics commonly used in the field of remote sensing classification are adopted to evaluate classification performance. OA measures the overall classification accuracy of all test samples. AA computes the arithmetic mean of the classification accuracy of each class. It reflects the balanced recognition ability for each class. The Kappa coefficient evaluates the consistency between the classification results and the ground truth labels.
2) *Experimental Environment:* All experiments are implemented based on the PyTorch deep learning framework. The hardware platform is configured with an Intel Core i5-13400 processor and an NVIDIA GeForce RTX 4070 graphics card.
3) *Hyperparameter Settings:* The total number of training epochs is set to 100. The batch size is 64. The Adam optimizer is adopted, with an initial learning rate of 0.001 and a weight decay coefficient of 0.0005. The learning rate decays by a factor of 0.5 every 50 epochs. The loss function is cross-entropy loss.
4) *Training/Test Split Ratios:* Experiments are conducted on three benchmark datasets: Houston2013, Trento, and MUUFL. The training and test sample splits for each dataset are shown in Tables I–III, respectively.

C. Comparative Experiments

To evaluate the effectiveness of the proposed method, nine representative HSI-LiDAR joint classification methods are selected for comparison: MCFNet [21], IP-CNN [16], CrossHL [25], MTNet [24], GLT-Net [34], MADNet [22], CITNet [38], IACF-Net [41], and FETNet [32]. All methods adopt the parameter configurations consistent with their original papers. Evaluations are conducted on the Houston2013, Trento, and MUUFL datasets, respectively. Tables IV–VI present the classification results of each method on the Houston2013, Trento, and MUUFL datasets. The classification maps of each method on the three datasets are visualized in Figs. 8–10. The figures also show the false-color image and the ground truth map.

As shown in Table IV and Fig. 8, the proposed method achieves the highest OA of 95.18%, outperforming IACF-Net and FETNet, which rank second and third with 94.58% and 93.12%, respectively. On confusing classes with similar spectra but different elevations, such as C10 Highway and C12 Parking Lot 1, MCESAN achieves 92.08% and 92.99%, far surpassing the second-best results of 79.73% and 92.32%. This indicates that ADPM effectively captures local spatial discrepancies and SKSA utilizes elevation differences to separate spectrally similar categories.

As shown in Table V and Fig. 9, MCESAN achieves the best results with an OA of 99.21%, compared to IACF-Net at 98.67% and FETNet at 98.52%. On the easily confused classes C3 Ground and C6 Roads, MCESAN reaches 98.66% and 94.43%, improving by about 6%–8% over the second-best method. Here, GCCM's global semantic calibration proves particularly effective by suppressing noisy spectral channels and enhancing elevation-related information.

As shown in Table VI and Fig. 10, MCESAN achieves the highest OA of 92.95%, AA of 92.38%, and Kappa of 0.9063, consistently outperforming IACF-Net and FETNet. On fine-grained and imbalanced categories such as C1 Trees and C3 Mixed ground surface, the improvements are especially significant. The SKSA mechanism prevents redundant associations from diluting attention weights in complex urban scenes, maintaining stable discrimination capability under high interference.

In summary, MCESAN maintains its lead across all three datasets. Houston2013 confirms its ability to distinguish similar objects, Trento demonstrates its stability in vegetation-dominated scenes, and MUUFL proves its robustness under class imbalance and fine-grained categories. The performance gains stem from two key factors: MCEC eliminates modality differences from both local and global dimensions, while SKSA focuses on key cross-modal dependencies through sparse selection. Their synergistic cooperation enables MCESAN to outperform all compared methods across multiple evaluation dimensions.

D. Parameter Experiment

To determine the optimal configuration of key hyperparameters, the effects of the input patch size, the sparsity parameter

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TABLE IV

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE HOUSTON2013 DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class No.	MCFNet	IP-CNN	Cross-HL	MTNet	GLTNet	MADNet	CITNet	IACF-Net	FETNet	MCESAN
C1	89.74	**94.49**	84.23	82.91	94.30	83.10	82.72	95.12	92.87	87.27
C2	98.68	97.09	98.97	97.37	98.50	99.62	**100.00**	99.78	99.45	99.91
C3	98.61	98.42	85.15	87.52	99.41	**99.60**	97.43	99.52	98.96	97.82
C4	98.96	89.02	**99.24**	97.03	97.35	89.87	97.16	99.01	98.33	98.86
C5	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
C6	**100.00**	90.21	**100.00**	97.90	83.92	98.60	97.20	100.00	99.21	**100.00**
C7	82.46	77.61	90.76	92.63	76.40	88.90	84.79	92.18	90.23	**93.75**
C8	93.35	78.35	88.70	88.98	84.90	94.30	93.83	**94.87**	93.55	93.73
C9	78.47	87.63	92.73	81.87	75.92	97.83	98.58	**98.92**	97.01	94.33
C10	80.79	55.02	69.40	81.95	75.58	79.73	53.96	91.03	88.67	**92.08**
C11	93.45	98.77	95.63	91.75	**100.00**	88.14	93.74	98.02	97.55	95.16
C12	92.18	79.73	83.96	75.41	77.62	92.32	90.78	**93.85**	91.44	92.99
C13	81.40	84.91	91.93	83.51	**92.98**	87.02	90.18	91.25	90.76	89.47
C14	**100.00**	**100.00**	98.38	96.76	**100.00**	**100.00**	**100.00**	99.68	**100.00**	99.60
C15	**100.00**	**100.00**	97.67	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
OA(%)	91.65	87.22	90.79	89.72	89.32	92.24	90.66	94.58	93.12	**95.18**
AA(%)	92.61	88.75	91.78	90.52	90.46	93.27	92.02	95.21	94.33	**95.67**
κ *100	90.84	86.12	89.99	88.84	88.40	91.57	89.86	94.22	92.78	**94.77**

TABLE V

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE TRENTO DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class No.	MCFNet	IP-CNN	Cross-HL	MTNet	GLTNet	MADNet	CITNet	IACF-Net	FETNet	MCESAN
C1	99.59	99.41	99.87	97.23	98.77	99.51	**99.95**	99.62	99.52	99.87
C2	99.78	99.78	99.50	99.50	99.93	99.93	99.78	**99.96**	99.82	99.10
C3	96.26	98.13	93.85	95.19	97.33	99.73	94.65	**99.28**	98.41	98.66
C4	**100.00**	**100.00**	99.97	99.13	**100.00**	**100.00**	99.93	99.75	100.00	99.91
C5	93.31	89.13	99.57	99.34	97.49	95.89	99.34	99.52	99.03	**99.82**
C6	90.66	91.38	82.99	86.60	90.30	91.15	92.46	93.52	92.55	**94.43**
OA(%)	96.56	95.17	97.93	97.63	97.91	97.56	98.87	98.67	98.52	**99.21**
AA(%)	96.60	96.31	95.96	96.16	97.30	97.70	97.69	98.12	97.82	**98.63**
κ *100	95.42	93.62	97.23	96.83	97.20	96.75	98.49	97.86	97.36	**98.94**

TABLE VI

CLASSIFICATION RESULTS OF DIFFERENT METHODS ON THE MUUFL DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Class No.	MCFNet	IP-CNN	Cross-HL	MTNet	GLTNet	MADNet	CITNet	IACF-Net	FETNet	MCESAN
C1	92.26	90.48	89.76	90.84	91.31	92.34	93.94	94.58	93.52	**96.54**
C2	82.96	89.88	86.99	72.89	68.42	**90.56**	79.54	89.82	88.96	84.08
C3	63.61	72.19	86.14	79.98	83.48	69.74	82.96	86.44	85.62	**86.88**
C4	85.02	93.62	96.78	82.70	97.43	**98.03**	93.56	97.05	97.12	96.00
C5	82.77	89.03	89.83	90.06	84.06	88.83	**91.31**	90.41	89.47	89.83
C6	97.78	99.37	98.73	99.68	98.73	99.68	**100.00**	99.74	99.41	99.05
C7	89.58	96.40	**96.69**	90.25	86.80	92.80	88.81	95.06	93.67	92.08
C8	94.45	95.06	91.25	96.01	96.11	96.45	96.01	**96.74**	95.94	95.88
C9	78.95	77.98	85.91	78.62	72.96	78.70	**86.15**	85.52	84.33	85.83
C10	96.97	96.97	**100.00**	93.94	90.91	96.97	93.94	98.22	97.52	90.91
C11	96.64	99.16	**100.00**	98.32	99.16	99.16	93.94	99.48	99.23	99.16
OA(%)	86.27	88.54	89.75	88.01	87.78	89.26	90.94	91.72	90.78	**92.95**
AA(%)	87.36	90.92	**92.92**	88.48	88.13	91.21	91.40	92.26	91.89	92.38
κ *100	82.06	85.04	86.62	84.27	84.01	85.92	88.05	89.58	88.87	**90.63**

k, and the number of Transformer encoder layers on classification performance are analyzed respectively. Experiments are conducted on the Houston2013, Trento, and MUUFL datasets. Only one parameter is adjusted each time, while the others are kept at their initial settings.

1) Patch Size Sensitivity Experiment: The patch size determines the local spatial neighborhood range of the model input samples. It directly affects the completeness of local features. Six patch sizes are selected for experiments: $7 \times 7, 9 \times 9, 11 \times 11, 13 \times 13, 15 \times 15$, and 17×17. The results are shown in Fig. 11.

The experimental results show that on all three datasets, OA first increases and then decreases as the patch size grows. It reaches the optimum at 13×13. A smaller patch lacks sufficient spatial context. The local structural information of ground objects is incomplete. An excessively large patch tends to introduce mixed pixels and background noise. This reduces feature purity. The Houston2013 and MUUFL scenes are complex, with interleaved object boundaries. The redundant interference caused by large patch sizes is more pronounced in these datasets. Therefore, the optimal patch size is set to 13×13.

2) Sparsity Parameter k Sensitivity Experiment: The parameter k controls the number of key interaction associations retained in each row of SKSA. It directly affects the precision of cross-modal attention filtering. Six values of k are tested: 0.4, 0.5, 0.6, 0.7, 0.8, and 0.9. The results are shown in Fig. 12.

The results show that OA first improves as k increases, then stabilizes or even slightly declines. The overall optimum is achieved at $k = 0.7$. When k is too small, the attention matrix

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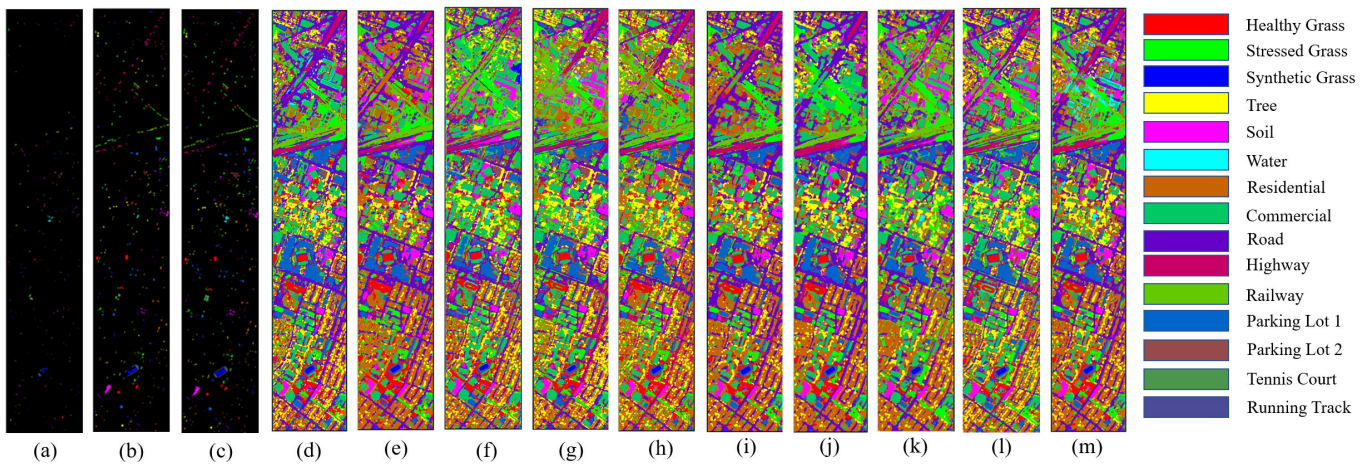


Fig. 8. Classification maps generated by different comparison methods for the Houston2013 dataset. (a) Distribution map of training samples. (b) Distribution map of testing samples. (c) Ground-truth map. (d) MCFNet. (e) IP-CNN. (f) Cross-HL. (g) MTNet. (h) GLTNet. (i) MADNet. (j) CITNet. (k) IACF-Net. (l) FETNet. (m) MCESAN.

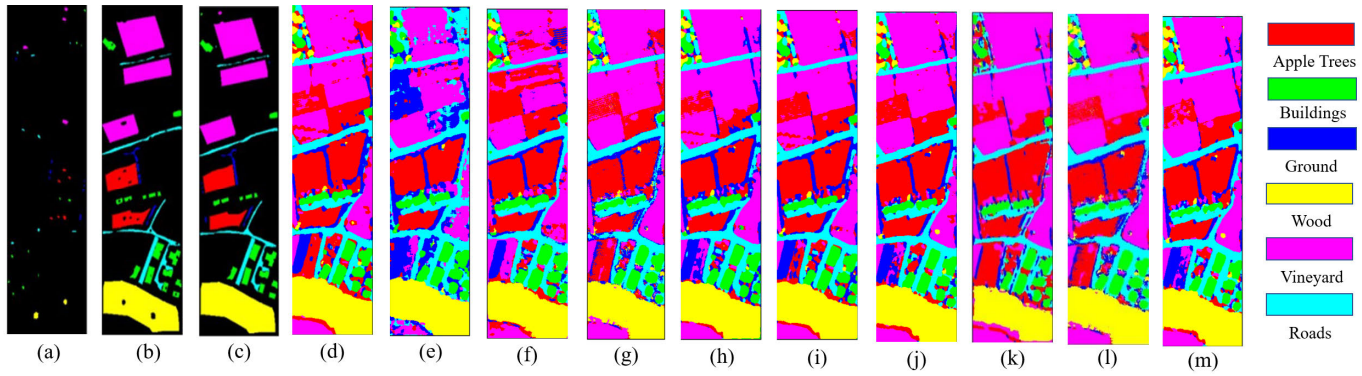


Fig. 9. Classification maps generated by different comparison methods for the Trento dataset. (a) Distribution map of training samples. (b) Distribution map of testing samples. (c) Ground-truth map. (d) MCFNet. (e) IP-CNN. (f) Cross-HL. (g) MTNet. (h) GLTNet. (i) MADNet. (j) CITNet. (k) IACF-Net. (l) FETNet. (m) MCESAN.

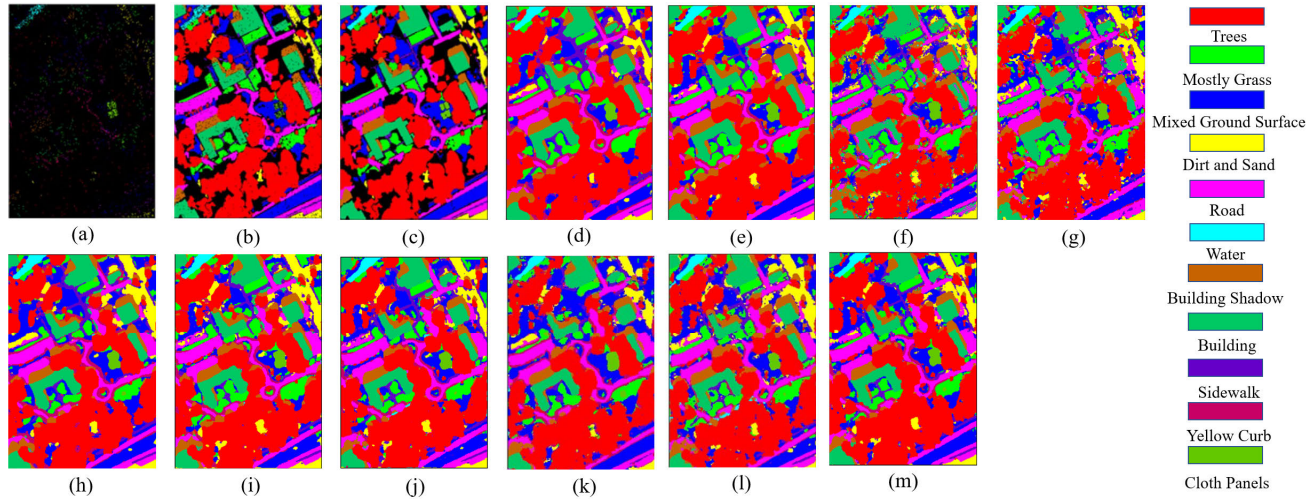


Fig. 10. Classification maps generated by different comparison methods for the MUUFL dataset. (a) Distribution map of training samples. (b) Distribution map of testing samples. (c) Ground-truth map. (d) MCFNet. (e) IP-CNN; (f) Cross-HL. (g) MTNet. (h) GLTNet. (i) MADNet. (j) CITNet. (k) IACF-Net. (l) FETNet. (m) MCESAN.

is over-compressed. Some effective cross-modal associations are discarded. This leads to insufficient information utilization. When k is too large, many low-contribution redundant interactions are retained. This dilutes the attention weights of key features. In complex scenes such as MUUFL, the performance drop with an excessively large k is more evident. This indicates that precise filtering is more necessary when the confusion among ground objects is high. The optimal k is set to 0.7.

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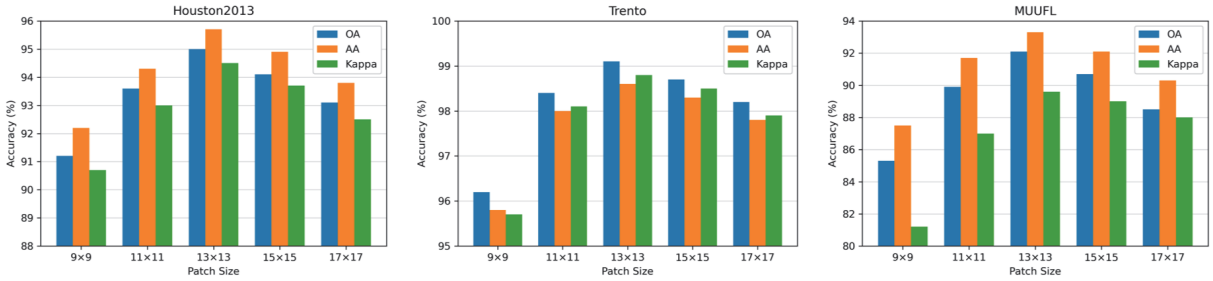


Fig. 11. Classification performance under different patch sizes.

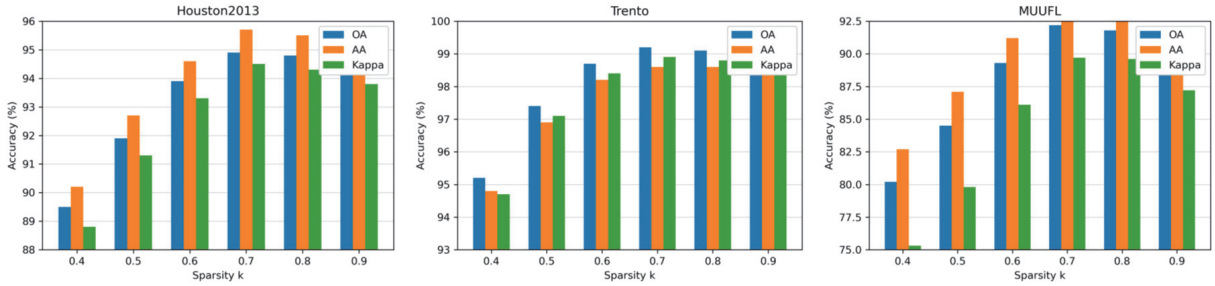


Fig. 12. Classification performance under different sparsity parameter k values.

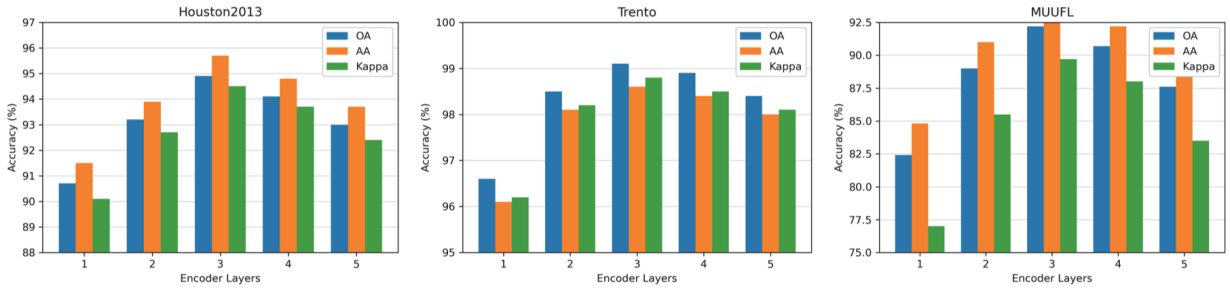


Fig. 13. Classification performance under different numbers of Transformer encoder layers.

3) Experiment on the Number of Transformer Encoder Layers: The number of encoder layers determines the modeling depth of long-range dependencies. The layers are tested from 1 to 5. The results are shown in Fig. 13.

The results show that on all three datasets, OA first increases and then decreases as the number of layers increases. The best performance is achieved with three layers. Too few layers cannot fully capture the complex global dependencies between HSI spectral features and LiDAR spatial features. Too many layers introduce the risk of overfitting and increase the computational cost. Houston2013 has many classes and high interclass confusion. The improvement is most significant when the number of layers increases from 2 to 3. This indicates that deep global modeling is more necessary for complex scenes. The optimal number of layers is set to 3.

Based on the results of the parameter analysis, the subsequent experiments uniformly adopt a patch size of 13×13, a sparsity parameter k of 0.7, and three Transformer encoder layers.

E. Ablation Experiment

To verify the effectiveness of each core module in the proposed method, ablation experiments are conducted on the three datasets: Houston2013, Trento, and MUUFL. The adaptive difference perception module, global consistency correction module, and SKSA are removed, respectively, to construct different model variants. The comparison is made using OA, AA, and the Kappa coefficient as metrics. The results are shown in Table VII.

After removing ADPM, OA drops by 1.90%, 0.83%, and 2.26% on the three datasets. ADPM eliminates modality misalignment through per-pixel discrepancy quantification and correction, with more significant effects on complex scenes like MUUFL and Houston2013, indicating that local spatial alignment is the foundation for subsequent global modeling.

After removing GCCM, OA drops by 2.17%, 1.05%, and 2.59%. GCCM characterizes channel semantic distributions through multipooling and achieves global semantic calibration via cross-modal channel weights. Its role is particularly critical on Houston2013 and MUUFL, which have more classes and complex semantic distributions.

After removing SKSA, the model degrades to traditional self-attention, with OA drops of 3.07%, 1.93%, and 3.04%—the largest among the three modules. Traditional self-attention indiscriminately computes all feature relationships, allowing redundant associations to dilute key cross-modal

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TABLE VII

CLASSIFICATION RESULTS OF DIFFERENT COMBINATIONS ON THE BENCHMARK DATASETS

Module			Metrics	Datasets		
AD PM	GC CM	SK SA		Houston 2013	Trento	MUUFL
×	×	✓	OA (%)	91.02	96.87	88.76
			AA (%)	92.35	96.52	89.43
			κ *100	90.52	96.43	85.31
×	✓	✓	OA (%)	93.28	98.38	90.69
			AA (%)	94.12	97.91	91.32
			κ *100	92.98	98.09	87.65
✓	×	✓	OA (%)	93.01	98.16	90.36
			AA (%)	93.89	97.69	90.95
			κ *100	92.65	97.88	87.21
✓	✓	×	OA (%)	92.11	97.28	89.91
			AA (%)	93.27	96.89	90.53
			κ *100	91.79	97.05	86.73
✓	✓	✓	OA (%)	95.18	99.21	92.95
			AA (%)	95.67	98.63	92.38
			κ *100	94.77	98.84	90.63

TABLE VIII

CLASSIFICATION PERFORMANCE UNDER DIFFERENT MISREGISTRATION LEVELS ON THREE DATASETS

Shift Level	Houston 2013		Trento		MUUFL	
	w/o ADPM	w/ ADPM	w/o ADPM	w/ ADPM	w/o ADPM	w/ ADPM
0px	93.28	**95.18**	98.38	**99.21**	90.69	**92.95**
0.5px	92.01	94.56	97.86	98.97	89.37	92.16
1px	90.72	93.87	97.12	98.54	88.04	91.28
1.5px	89.14	92.95	96.25	98.02	86.48	90.14
2px	87.56	91.76	95.08	97.38	84.76	88.89
3px	84.58	89.52	93.42	96.21	81.93	86.77

dependencies. SKSA constructs asymmetric interaction with LiDAR as query and HSI as key/value, while the sparse selection operator purifies key associations. The improvement from SKSA is most significant on datasets with high class confusion.

To verify the synergistic effect between ADPM and GCCM, we also removed both simultaneously while keeping only SKSA. The results in Table VII show that on Houston2013, OA drops to 91.02%, exceeding the sum of their individual declines of 1.90% and 2.17%. This confirms that ADPM and GCCM jointly provide a high-quality feature foundation for SKSA, with their cooperation contributing beyond individual contributions.

To further evaluate ADPM's robustness against spatial misregistration, we conducted controlled experiments on all three datasets by artificially introducing horizontal and vertical shifts to the LiDAR data at six levels: 0, 0.5, 1, 1.5, 2, and 3 pixels. For each shift level, we compared the full MCESAN with a variant where ADPM was removed. The results are presented in Table VIII.

When perfectly aligned, MCESAN with ADPM consistently outperforms the variant without ADPM across all three

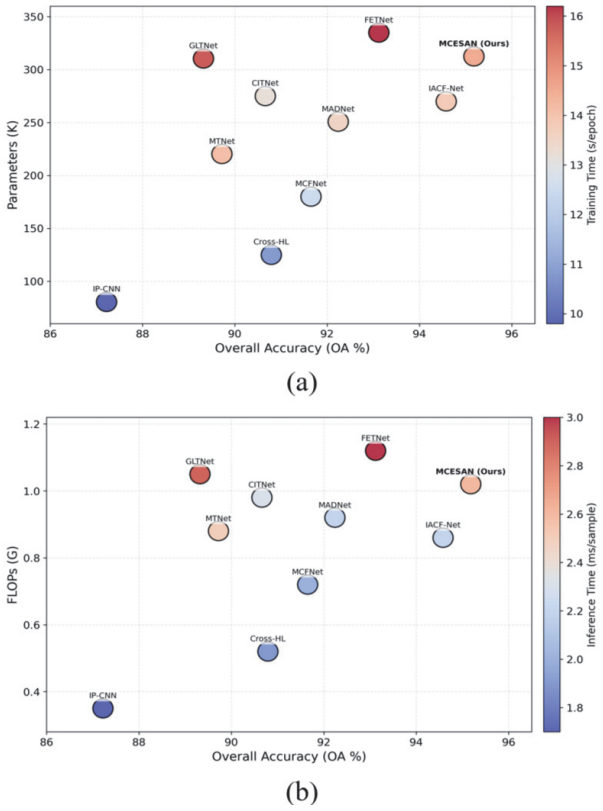


Fig. 14. Comparison of classification accuracy and computational cost for different methods. (a) OA versus number of parameters. (b) OA versus FLOPs.

datasets. As misregistration increases, both variants degrade, but the variant with ADPM consistently exhibits slower degradation and maintains a clear advantage. At a three-pixel shift, MCESAN with ADPM outperforms the variant without ADPM by 4.94, 2.79, and 4.84 percentage points on Houston2013, Trento, and MUUFL, respectively, with the gap widening as shift increases. These results confirm ADPM's effectiveness in mitigating misregistration across diverse scenes.

From the comprehensive ablation results, all three modules contribute positively to classification performance, with SKSA making the greatest contribution. ADPM and GCCM provide a high-quality feature foundation from local spatial and global channel dimensions, respectively, and the misregistration experiments further demonstrate ADPM's robustness under imperfect alignment across all three datasets. The three modules work together to achieve optimal performance.

F. Efficiency Analysis

To comprehensively evaluate the practical deployment potential of MCESAN, this article quantitatively analyzes the computational efficiency of the model from four dimensions: number of parameters (Params), floating-point operations (FLOPs), training time per epoch, and inference time per sample. The input patch size is uniformly set to 13 × 13, and all metrics are measured under the same environment on the Houston2013 dataset. We compare MCESAN with nine representative methods: MCFNet [21], IP-CNN [16], Cross-HL [25], MTNet [24], GLT-Net [34], MADNet [22], CITNet [38], IACF-Net [41], and FETNet [32].

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Fig. 14 presents the tradeoff between classification accuracy and computational cost for each model in the form of bubble charts. Fig. 14(a) shows OA versus the number of parameters, while (b) shows OA versus FLOPs. The bubble color in each subfigure indicates the training or inference time of the corresponding method.

IP-CNN has the lowest number of parameters and the fastest inference speed. However, its OA is only 87.22%, far lower than that of other methods. Cross-HL also has relatively high computational efficiency, but its OA is 90.79%, still unsatisfactory. GLTNet has a large number of parameters, 310.5 K, and FLOPs of 1.05 G, yet its OA is only 89.32%, showing poor performance in the accuracy-efficiency tradeoff. IACF-Net achieves a strong OA of 94.58% with 270.0 K parameters and 0.86 G FLOPs, demonstrating the best accuracy-efficiency balance among the compared methods except MCESAN. FET-Net obtains an OA of 93.12% but incurs higher computational cost with 335.0 K parameters and 1.12G FLOPs.

In contrast, MCESAN achieves the highest OA of 95.18%, significantly outperforming all compared methods. It contains 312.5 K parameters and requires 1.02 G FLOPs, with a training time of 14.5 s per epoch and an inference time of 2.6 ms per sample. The parameter count is slightly higher than that of IACF-Net, but the FLOPs are comparable; both parameters and FLOPs are lower than those of FETNet. This indicates that the SKSA mechanism effectively controls the computational complexity in the global modeling stage through the sparse selection operator. Overall, MCESAN achieves a significant accuracy improvement with an acceptable increase in computational cost, reaching the best balance among all nine compared methods.

IV. Conclusion

Targeting insufficient bimodal feature consistency and redundant dependencies in long-range modeling for joint HSI and LiDAR classification, this article proposes a method based on modality complementary enhancement and sparse long-range modeling. An MCEC module, consisting of an adaptive difference perception module and a global consistency correction module, achieves precise alignment and semantic calibration of bimodal features in both local space and global channel dimensions. This effectively eliminates modality misalignment and noise from different imaging mechanisms. On this basis, a SKSA mechanism is designed. It builds a cross-modal directed interaction framework with LiDAR spatial features as queries and HSI spectral features as keys and values. A sparse selection operator adaptively refines the attention similarity matrix, pre-serving key cross-modal dependencies while suppressing the dilution of attention weights by low-contribution redundant associations.

Experimental results on three public benchmark datasets demonstrate that the proposed method achieves state-of-the-art performance, significantly outperforming nine representative comparison methods across all evaluation metrics. These results verify the method's effectiveness and robust-ness for high-accuracy land-cover classification in complex scenes. Future research directions include: exploring dynamic sparse strategies based on feature distribution statistics or learnable gating mechanisms to automatically adjust filtering strength; studying a unified correction framework with joint local–global optimization to further enhance bimodal feature consistency; investigating cross-region generalization and few-shot scenarios; reducing computational cost through model pruning or knowledge distillation while maintaining accuracy; and extending the method to broader remote sensing applications.

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